

A. RAMCHARAN

Aspiring AI/ML Engineer | RAG & Full-Stack Application Builder

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PROFESSIONAL SUMMARY

Aspiring AI/ML engineer with hands-on experience building and deploying Retrieval-Augmented Generation (RAG) systems using the OpenAI API — including document chunking, embedding-based retrieval, and grounding LLM responses in custom knowledge bases to reduce hallucinations. Full-stack foundation in React.js, Node.js, Express.js, and MongoDB, with practical experience turning raw, messy data into structured inputs for production apps. Comfortable judging when an LLM/RAG pipeline is the right tool versus a simpler rule-based approach, and explaining that reasoning to non-technical stakeholders. Solid grounding in DSA, OOP, DBMS, and OS.

RELEVANT AI/ML & RAG EXPERIENCE

- Built and deployed a production RAG chatbot on top of the OpenAI API: cleaned and preprocessed source documents, chunked content, generated embeddings, and implemented semantic retrieval so responses were grounded in a custom knowledge base rather than the model's general training data.
- Reduced irrelevant/hallucinated responses by tuning retrieval relevance (chunk size, top-k) rather than relying on prompt engineering alone — a deliberate choice, not a default to RAG.
- Can articulate when RAG is the wrong tool: for a small, static FAQ set, a direct prompt or cached lookup beats RAG on latency and cost, since retrieval overhead isn't justified when the knowledge base is tiny and rarely changes.
- Practiced explaining RAG architecture (retrieval + generation, why grounding reduces hallucination) in plain language to non-technical users during project demos.

WORK EXPERIENCE

Full Stack Development Trainee | NxtWave CCBP 4.0 Academy

2024 – Present

- Engineered full-stack web applications end-to-end using React.js, Node.js, Express.js, and MongoDB, following MVC architecture and RESTful API design principles.
- Designed and implemented secure REST APIs with JWT-based authentication and role-based access control, tightening access control across protected routes.
- Built and deployed an AI-powered chatbot leveraging OpenAI APIs and Retrieval-Augmented Generation (RAG), including cleaning and structuring the underlying knowledge base to deliver accurate, low-latency, context-aware responses.
- Solved 150+ Data Structures & Algorithms problems and collaborated using Git/GitHub for version control across all projects.

PROJECTS

EduReach — Agentic AI College Chatbot Platform

Live Demo: [edureach-platform-vert.vercel.app](#) | Stack: React.js, Node.js, Express.js, MongoDB, JWT, OpenAI API, RAG

- Architected and shipped a full-stack, agentic AI chatbot platform for college students, deployed live on Vercel.
- Built a RAG pipeline (data cleaning, chunking, embedding generation, semantic retrieval) with the OpenAI API to ground chatbot responses in a custom institutional knowledge base, cutting down irrelevant and hallucinated replies.
- Implemented JWT authentication and RBAC for differentiated student/administrator access, and designed RESTful APIs with a responsive, mobile-friendly UI for real-time interactions.

Restaurant QR Ordering System — Multi-Panel Ordering Platform

Live Demos: [Admin Panel](#) | [Kitchen Panel](#) | [Customer Web App](#) | Stack: React.js, Node.js, Express.js, MongoDB, REST APIs, Real-time Order Sync

- Designed and built a QR-code-based contactless ordering system with three independently deployed panels — Customer Web App, Kitchen Display, and Admin Dashboard — sharing a unified MongoDB backend.
- Developed a customer-facing ordering interface via per-table QR codes and a real-time Kitchen Panel that displays incoming orders instantly, streamlining preparation and cutting manual order-taking errors.
- Created an Admin Panel for menu management, order tracking, and analytics; engineered RESTful APIs to keep order state synchronized across all three panels in real time.

TECHNICAL SKILLS

- AI/ML: OpenAI API, Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, Semantic Search
- Languages: Python, JavaScript (ES6+), SQL, HTML5, CSS3
- Backend: Node.js, Express.js, REST API Design, JWT Authentication
- Frontend: React.js, Bootstrap, Responsive Web Design
- Database: MongoDB, MySQL | Tools & Platforms: Git, GitHub, VS Code, Vercel, Docker (familiar)
- Core CS Fundamentals: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

EDUCATION

- B.Tech, Computer Science & Engineering — JBIET College, Expected 2027, CPI: 8.44
- Intermediate (MPC) — Saraswathi Junior College, 2023, CPI: 9.70
- SSC — Green Grove High School, 2020, CPI: 10.0